Villa Complex 415V LV Network

Electrical design report

Project	Villa Complex 415V LV Network
Project no.	*Not defined*
Client	Mr. Ahmed Al Mansoori
Consultant	ABC Engineering Consultants
Contractor	*Not defined*
Plot / area	123-456 · VILLA, UAE
Scope	Whole installation (5 boards)
Studies	Load assessment; Transformer and standby generator sizing; LV distribution; Cable and breaker sizing; Load flow and voltage drop study; Short circuit study; Protection discrimination study; Earth fault loop impedance and disconnection; Phase balance; Busbar trunking risers; Power factor correction; DB and cable schedules
System	415 V, 3-phase + N, 50 Hz, TN-S
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	Name	Signature	Date
Prepared by	QA		
Checked by	*Not defined*		
Approved by	*Not defined*		

Generated by LV Design Studio from the latest calculation run. Calculations cover the whole network; this report shows the part in scope. Calculated checks against the project limits — engineering review and approval are separate. Results must be checked by a qualified engineer before submission. **Cable data — 2.5, 4, 6, 10, 16, 25, 35, 50, 95, 150, 300 mm²: app reference value (approximate, not manufacturer-verified). Verify against the manufacturer's datasheet before relying on these results.**

Document control

Document information

Document title	Electrical design report
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Revision history

> No revision has been issued for this project (Revisions → Issue revision). This report is an unissued draft.

Prepared, checked and approved

Role	Name	Signature	Date
Prepared by	QA		
Checked by	*Not defined*		
Approved by	*Not defined*		

Calculation record

What these results were calculated from, with what, and how far they can be relied on.

Project	Villa Complex 415V LV Network Project ID not assigned (save the project) · file format version 1
Calculated	At the time of this print, from the current inputs
Software	LV Design Studio 1.38.0 · built-in calculation engine (no external engine used)
Calculation basis	Methods per Codes and standards → Calculation basis and each study's Method notes. Standard editions: not verified against a specific edition
Key assumptions	• 415 V, 3-phase + N, 50 Hz, TN-S · design ambient 45 °C · voltage drop limit 4 % source to load • Cable resistance at operating temperature 1.2 × R20 (not set for the project) • Fault levels: maximum Ik″ with c max = 1.05 and cable R at 20 °C, MV network infinite, transformer correction KT not applied — **simplified, preliminary, not a full IEC 60909 study**
Cable data	**11 sizes used: 11 app reference values (approximate, not manufacturer-verified) · 0 manufacturer data** Final-design requirement for manufacturer cable data: not set
Input checks	No invalid inputs, no supply loops, no missing references
Calculation warnings	0 failures · 34 to check (see the Results summary under Contents)
Limitations	• Cable ratings: per installation method with ambient / ground, soil and grouping factors; grouping with other circuits is counted only on cable tray routes • Selectivity from current thresholds only — confirm with the manufacturer's tables • Voltage drop: first-order, summed along the radial path; not an unbalanced load flow

Validation status: PRELIMINARY. Calculated checks against the project limits, using approximate reference or unverified cable data. Not an engineering approval and not an approval by DEWA or any other authority.

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Results summary

Section	Study	Result
6	Load assessment	—
7	Transformer and standby generator sizing	2 pass · 0 warning · 0 fail
8	LV distribution	9 pass · 0 warning · 0 fail
9	Cable and breaker sizing	36 pass · 0 warning · 0 fail
10	Load flow and voltage drop study	22 pass · 0 warning · 0 fail
11	Short circuit study	36 pass · 0 warning · 0 fail
12	Protection discrimination study	0 pass · 31 warning · 0 fail
13	Earth fault loop impedance and disconnection	36 pass · 0 warning · 0 fail
14	Phase balance	3 pass · 2 warning · 0 fail
15	Busbar trunking risers	—
16	Power factor correction	0 pass · 1 warning · 0 fail
17	DB and cable schedules	—

1. Executive summary

Connected load	Maximum demand	Demand ÷ connected	Power factor · current
514 kW	**451 kW · 512 kVA**	**0.88**	**0.88 · 712 A**

1.1 Transformer requirement

Table 1

Main board	Required (kVA)	Recommended	Installed (kVA)	Loading	Status
MDB-1	710	1000 kVA	1,000	47 %	PASS

1.2 Generator requirement

Table 2

Essential demand (kVA)	Rating needed (kVA)	Governed by	Recommended	Installed (kVA)	Status
64	544	motor start	630 kVA / 504 kW	*Not defined*	NOT CHECKED

1.3 Main LV distribution

- MDB-1 (MDB) feeding 4 board(s): 2 SMDB, 1 DB, 1 MCC

1.4 Major design findings

Table 3

Status	Finding
PASS	Transformer and standby generator sizing: all 2 checks pass.
PASS	LV distribution: all 9 checks pass.
PASS	Cable and breaker sizing: all 36 checks pass.
PASS	Load flow and voltage drop study: all 22 checks pass.
PASS	Short circuit study: all 36 checks pass.
WARNING	Protection discrimination study: 31 check(s) need review.
PASS	Earth fault loop impedance and disconnection: all 36 checks pass.
WARNING	Phase balance: 2 check(s) need review.
WARNING	Power factor correction: 1 check(s) need review.

Whole installation, from the latest calculation run. Section results are for the scope of this report.

2. Project scope

> The project scope is not defined (Project → Brief). Included and excluded systems are not listed.

2.1 Electrical systems included

Not defined

2.2 Electrical systems excluded

Not defined

2.3 Design boundaries

- Starts at: the LV terminals of the transformer feeding MDB-1.
- MV network upstream of the transformers is outside the calculations (treated as an infinite source).
- Ends at: the final circuits on the DB load schedules and the loads connected to them.

3. Electrical system description

3.1 Incoming supply

- MDB-1: from a dedicated transformer

3.2 MV / LV arrangement and transformers

- MDB-1: 1000 kVA, 5 % Z, Dyn11 (app default), 415 V secondary

3.3 Main LV distribution

- 415 V, 3-phase + N, 50 Hz, TN-S (the earthing arrangement the calculations assume).
- MDB-1 (MDB) feeding 4 board(s): 2 SMDB, 1 DB, 1 MCC

3.4 Emergency supply

None defined in the project.

3.5 UPS

None defined in the project.

4. Codes and standards

4.1 Project and authority requirements

Approving authority	*Not defined*

> No project standards or specifications are defined (Reports → Study reports → Output). None are assumed.

4.2 Calculation basis

Table 4

Standard	Used for
IEC 60909-0	Short-circuit currents (simplified: c factor, infinite MV network)
IEC 60364-5-52	Cable current-carrying capacity, derating and voltage drop
IEC 60364-4-43	Overload protection: Ib ≤ In ≤ Iz
IEC 60364-4-41	Automatic disconnection: earth fault loop impedance and disconnection times
IEC 60364-5-54	Protective conductor size (Table 54.2 / adiabatic)
IEC 60898-1 / IEC 60947-2	Circuit-breaker characteristics and breaking capacity
IEC 60831	Capacitor bank rating and switching device
IEC 61439-6	Busbar trunking ratings

The methods the app's calculations follow for the studies in this report. They are not a statement of the project's contractual standards.

5. Design criteria

Table 5

Parameter	Design value	Source	Note
System voltage (line-line)	415 V	Project settings	
Frequency	50 Hz	Project settings	
Earthing arrangement	TN-S	Calculation basis (fixed)	
Design ambient temperature	45 °C	Project settings	
Voltage drop limit (source to load)	4 %	Project settings	
Conductor temperature for voltage drop	1.2 × R20	Calculation basis	*Not set for the project*
Final circuit disconnection	IEC 60364-4-41 Table 41.1	Project settings	
Power factor target	0.95	App default	*Not set for the project — app default used*
Future growth allowance (transformer)	20 %	App default	*Not set for the project — app default used*
Transformer maximum design loading	80 %	App default	*Not set for the project — app default used*
Generator maximum design loading	80 %	App default	*Not set for the project — app default used*
Maximum demand factor (authority forms)	0.8	Project information	
Overall demand ÷ connected load	0.88 (451 ÷ 514 kW)	Load assessment (calculated)	

6. Load assessment

Scope: Whole installation (5 boards)

6.1 Input

Loads	Connected loads on the feeders and DB load schedules
Demand factors	Per load / circuit as entered

6.2 Design criteria

Future growth allowance	20 % (app default)
Transformer design loading	80 %

6.3 Method

- Demand = connected load × demand factor for each load, summed up the network (kW and kvar separately); maximum demand kVA = $\sqrt{(kW^2 + kvar^2)}$.
- Required capacity = maximum demand × (1 + growth) ÷ design loading, from the transformer sizing (no further diversity applied).

6.4 Result

Connected load
514 kW

Maximum demand
451 kW

Demand ÷ connected
0.88

6.5 Load at each board

Table 6

Board	Connected (kW)	Demand (kW)	Demand ÷ connected	Demand (kVA)	PF	Current (A)
MDB-1	514	451.1	0.88	512.1	0.88	712
SMDB-GF	231	203.3	0.88	233.7	0.87	325
DB-GF1	38	38	1	42.2	0.9	59
SMDB-FF	109	87	0.8	97.4	0.89	135
MCC-1	130	130	1	151.2	0.86	210

6.6 Design load of each main board

Table 7

Main board	Maximum demand (kVA)	Future growth	Required capacity (kVA)
MDB-1	473	20 %	710

7. Transformer and standby generator sizing

Scope: Whole installation (5 boards)

Overall status: **PASS** (2 pass · 0 warning · 0 fail)

7.1 Design criteria

Future growth	20 %
Transformer design loading	≤ 80 %
Generator design loading	≤ 80 %

7.2 Method

- Transformer per main board: maximum demand (after the planned power factor correction) × (1 + 20 % growth) ÷ 80 % design loading, next DEWA standard size (500 / 1000 / 1500 kVA).
- For that size: full-load current and main breaker, LV fault level (typical IEC 60076-5 impedance, infinite MV source), voltage regulation at the design demand.
- Bus couplers: with one transformer out, the other carries both boards up to 100 % of its rating. Duty / standby: two transformers, each for the whole load.
- Standby generator from the boards on it (share of each), plus circuits marked essential. The set must carry the running kVA and the running kW at 80 % loading: rating ≥ max(kVA ÷ 0.8, kW ÷ (0.8 × 0.8)), with rated kW = 0.8 × rated kVA unless the Generators list gives the set's kW; or larger if the largest motor, started last, would dip the voltage over 15 % (X'd 25 %).

7.3 Result

Items checked
2 pass · 0 warning · 0 fail

MDB-1 transformer
1000 kVA

Standby generator
630 kVA / 504 kW

7.4 Transformers

Table 8

Main board	Demand (kVA)	PF	Design (kVA)	Recommended	Installed · loading	FLC · main breaker	LV fault (kA) · lowest Icu	Regulation	Result
MDB-1	473	0.95	710	1000 kVA	1000 kVA · 47 %	1391 A · 1600 A	29.2 · 36	1.1 %	PASS

7.5 Standby generator — size

Table 9

Item	Value
Running demand	55 kW · 64 kVA
For the running load	86 kVA — running; the motor start sets the size
Largest motor start	MCC-FP: 384 kVA starting with 0 kVA already running
For the motor start	544 kVA (dip 13.2 % on the recommended set)
Recommended	630 kVA / 504 kW at 0.8 PF
Result	PASS
With a soft starter on the largest motor	300 kVA

Item	Value
Full-load current · ATS / breaker	876 A · 1000 A
Installed	—

7.6 Verification — Capacity verification

Table 10

Equipment	Calculated demand (kVA)	Required capacity (kVA)	Selected	Installed (kVA)	Loading	Spare (kVA)	Status
Transformer MDB-1	473	710	1000 kVA	1000	47 %	527	PASS
Standby generator	64	544	630 kVA	—	—	—	NOT CHECKED

See also Appendix D (equipment data).

8. LV distribution

Scope: Whole installation (5 boards)

Overall status: **PASS** (9 pass · 0 warning · 0 fail)

8.1 Input

Supply	MDB-1: transformer 1000 kVA, 5 % Z

8.2 Design criteria

Board loading	Demand current ≤ busbar / main device rating
Feeders	Ib ≤ In ≤ Iz

8.3 Method

- Board demand from the load flow; loading = demand current ÷ the board's rated current (when entered).

8.4 Result

Boards	Major feeders	Checks
5	**4**	**9 / 0 / 0 (pass / check / fail)**

8.5 Distribution boards

Table 11

Board	Type	Fed from	Bus rating (A)	Demand (A)	Loading	Ik″ (kA)
MDB-1	MDB	Supply	1600	712	45 %	29.2
SMDB-GF	SMDB	MDB-1	630	325	52 %	21.8
DB-GF1	DB	SMDB-GF	100	59	59 %	10.7
SMDB-FF	SMDB	MDB-1	250	135	54 %	14.3
MCC-1	MCC	MDB-1	400	210	53 %	19

8.6 Major feeders

Table 12

From	To	Breaker	Cable	Ib (A)	Iz (A)	Status
MDB-1	SMDB-GF	500 A MCCB	4C × 300 mm²	327	525	PASS
MDB-1	SMDB-FF	200 A MCCB	4C × 95 mm²	135	266	PASS
MDB-1	MCC-1	315 A MCCB	4C × 150 mm²	210	342	PASS
SMDB-GF	DB-GF1	100 A MCCB	4C × 35 mm²	60	146	PASS

9. Cable and breaker sizing

Scope: Whole installation (5 boards)

Overall status: **PASS** (36 pass · 0 warning · 0 fail)

9.1 Input

Cable	Cu / XLPE multicore, rated for each circuit's installation method (IEC 60364-5-52)
Ambient temperature	45 °C
Ground (buried cables)	20 °C, soil 2.5 K·m/W (IEC reference where not set)
Grouping	From the cable tray route (method E), else the circuit's own parallel runs

9.2 Design criteria

Overload protection	Ib ≤ In ≤ Iz (IEC 60364-4-43)
Voltage drop limit	4 %
Breaking capacity	Icu ≥ Ik″ at the breaker

9.3 Method

- Cu/XLPE/SWA multicore cable, rated for each circuit's IEC 60364-5-52 installation method (E free air / tray unless set); derated for air ambient (methods A–E) or ground temperature and soil thermal resistivity (D1 / D2, reference 20 °C and 2.5 K·m/W), and for grouping (the cable tray route for method E, else the circuit's own parallel runs: bunched for A–C, in ducts / in the ground for D1 / D2).
- Overload protection per IEC 60364-4-43: Ib ≤ In ≤ Iz. Voltage drop from the main board to the load within the limit. Breaker Icu ≥ the fault at its terminals.

9.4 Result

Circuits checked
36 pass · 0 warning · 0 fail

Ambient
45 °C

9.5 Selection — major feeders

Feeder: MDB-1 → SMDB-GF

Design current (Ib)	327 A
Protective device (In)	500 A MCCB
Selected cable	4C × 300 mm²
Installation	E · derating 0.87 · 1.00
Derated capacity (Iz)	525 A
Ib ≤ In ≤ Iz	PASS
Voltage drop (total)	0.46 % (limit 4 %)
Breaking capacity check	PASS
Overall status	PASS

Feeder: MDB-1 → SMDB-FF

Design current (Ib)	135 A
Protective device (In)	200 A MCCB
Selected cable	4C × 95 mm²
Installation	E · derating 0.87 · 1.00
Derated capacity (Iz)	266 A
Ib ≤ In ≤ Iz	PASS
Voltage drop (total)	0.75 % (limit 4 %)
Breaking capacity check	PASS
Overall status	PASS

Feeder: MDB-1 → MCC-1

Design current (Ib)	210 A
Protective device (In)	315 A MCCB
Selected cable	4C × 150 mm²
Installation	E · derating 0.87 · 1.00
Derated capacity (Iz)	342 A
Ib ≤ In ≤ Iz	PASS
Voltage drop (total)	0.58 % (limit 4 %)
Breaking capacity check	PASS
Overall status	PASS

Feeder: SMDB-GF → DB-GF1

Design current (Ib)	60 A
Protective device (In)	100 A MCCB
Selected cable	4C × 35 mm²
Installation	E · derating 0.87 · 1.00
Derated capacity (Iz)	146 A
Ib ≤ In ≤ Iz	PASS
Voltage drop (total)	0.92 % (limit 4 %)
Breaking capacity check	PASS
Overall status	PASS

9.6 Verification — Circuits on each board

Table 13

Board	Circuits	Pass / check / fail	Status
MDB-1	5	5 / 0 / 0	PASS
SMDB-GF	4	4 / 0 / 0	PASS
DB-GF1	21	21 / 0 / 0	PASS
SMDB-FF	4	4 / 0 / 0	PASS
MCC-1	2	2 / 0 / 0	PASS

See also Appendix B (cable calculations).

10. Load flow and voltage drop study

Scope: Whole installation (5 boards)

Overall status: **PASS** (22 pass · 0 warning · 0 fail)

10.1 Input

Supply	MDB-1: transformer 1000 kVA, 5 % Z
Loads	Design loads × demand factors (load schedules and feeders)

10.2 Design criteria

Voltage drop limit (source to load)	4 %

10.3 Method

- Balanced 3-phase load flow from the design loads: demand = connected load × demand factor, summed up the network (kW and kVAr separately).
- Voltage drop per cable: $\sqrt{3} \cdot I \cdot L \cdot (R \cos\phi + X \sin\phi)$ for 3-phase, $2 \cdot I \cdot L \cdot (\ldots)$ for single-phase; busbar voltage = nominal minus the drops of the incomers above it. Limit 4 % from the main board to the load.
- Transformer regulation is not included (busbar voltages are measured from the main LV busbar).
- Motors: drop while starting = upstream drop + cable drop × the starter's starting-current multiple; limit 10 %.

10.4 Result

Total demand of the scope
451 kW · 512 kVA

Lowest busbar voltage
99.08 %

Circuits checked
15 pass · 0 warning · 0 fail

10.5 Busbars

Table 14

Board	Connected (kW)	Demand (kW)	Demand (kVA)	PF	Current (A)	Rating (A)	Loading	Voltage (V)	% of nominal
MDB-1	514	451.1	512.1	0.88	712	1600	45 %	415	100
SMDB-GF	231	203.3	233.7	0.87	325	630	52 %	413.1	99.54
DB-GF1	38	38	42.2	0.9	59	100	59 %	411.2	99.08
SMDB-FF	109	87	97.4	0.89	135	250	54 %	411.9	99.25
MCC-1	130	130	151.2	0.86	210	400	53 %	412.6	99.42

10.6 Motor starting — limit 10 % source to motor

Table 15

Circuit	From	Motor	Running ΔV total (%)	Starting ΔV total (%)	Limit (%)	Result
MCC-WP	MCC-1	Motor (DOL)	1.03	3.3	10	PASS
MCC-FP	MCC-1	Fire pump (DOL)	1.12	3.8	10	PASS

10.7 Verification — Voltage drop — worst circuit on each board

Table 16

Board	Circuit	Calculated ΔV total (%)	Allowable (%)	Margin (%)	Status
MDB-1	DB-EV	2.38	4	1.62	PASS
SMDB-GF	GF-SKT	2.37	4	1.63	PASS
SMDB-FF	FF-LTG	2.57	4	1.43	PASS
MCC-1	MCC-FP	1.12	4	2.88	PASS

See also Appendix B (cable calculations).

11. Short circuit study

Scope: Whole installation (5 boards)

Overall status: **PASS** (36 pass · 0 warning · 0 fail)

11.1 Input

Supply	MDB-1: transformer 1000 kVA, 5 % Z
System	415 V, 50 Hz

11.2 Design criteria

Voltage factor	c max = 1.05 (maximum fault, cable R at 20 °C)
Breaking capacity	Breaker Icu ≥ prospective fault at its terminals (Ik″)

11.3 Method

- Maximum prospective symmetrical 3-phase fault current Ik″, simplified method based on IEC 60909-0 (preliminary — not a full IEC 60909 study): voltage factor c max = 1.05, cable R at 20 °C, upstream MV network infinite, transformer correction factor KT not applied.
- Source impedance from the transformer rating and %Z (split by its X/R ratio); cable R at 20 °C and X per km; parallel runs divide the impedance.
- Breaking capacity check: the fault at the breaker terminals (supply busbar) must not exceed the breaker Icu. Fault at the cable end: 3-phase for 3/4-core circuits, line-to-neutral for 2-core.
- Motor contribution and generation are neglected.

11.4 Result

Highest busbar fault
29.2 kA

Breakers checked
36 pass · 0 warning · 0 fail

11.5 Fault levels at the busbars

Table 17

Board	Type	Fed from	Busbar rating (A)	Ik″ max (kA)
MDB-1	MDB	Transformer 1000 kVA, 5 % Z	1600	29.2
SMDB-GF	SMDB	MDB-1	630	21.8
DB-GF1	DB	SMDB-GF	100	10.7
SMDB-FF	SMDB	MDB-1	250	14.3
MCC-1	MCC	MDB-1	400	19

11.6 Verification — Breaking capacity at each busbar

Table 18

Location	Ik″ busbar (kA)	Governing circuit	Ik″ at breaker (kA)	Icu (kA)	Margin (kA)	Breakers pass / check / fail	Status

Location	Ik″ busbar (kA)	Governing circuit	Ik″ at breaker (kA)	Icu (kA)	Margin (kA)	Breakers pass / check / fail	Status
MDB-1	29.2	INC-GF	29.2	36	6.8	5 / 0 / 0	PASS
SMDB-GF	21.8	GF-LTG	21.8	25	3.2	4 / 0 / 0	PASS
DB-GF1	10.7	DB-GF1-Y1	10.7	15	4.3	21 / 0 / 0	PASS
SMDB-FF	14.3	FF-OFF	14.3	25	10.7	4 / 0 / 0	PASS
MCC-1	19	MCC-WP	19	25	6	2 / 0 / 0	PASS

See also Appendix C (short-circuit and earth fault calculations).

12. Protection discrimination study

Scope: Whole installation (5 boards)

Overall status: **WARNING** (0 pass · 31 warning · 0 fail)

12.1 Design criteria

Overload selectivity	In upstream ÷ In downstream ≥ 1.6
Short-circuit selectivity	Fault at the downstream breaker below the upstream magnetic threshold

12.2 Method

- Each breaker against the breaker above it. Overload (time) selectivity: In upstream ÷ In downstream ≥ 1.6.
- Short-circuit selectivity: total when the maximum fault at the downstream breaker stays below the upstream magnetic no-trip threshold; otherwise partial up to that current. Confirm with the manufacturer's selectivity tables.

12.3 Result

Pairs checked

0 pass · 31 warning · 0 fail

12.4 Selectivity

Table 19

Upstream	Downstream	In ratio	Fault at downstream (kA)	Selectivity limit (kA)	Short circuit	Result
INC-GF (500 A)	GF-LTG (80 A)	6.25	21.8	4	Partial	WARNING
INC-GF (500 A)	GF-SKT (40 A)	12.5	21.8	4	Partial	WARNING
INC-GF (500 A)	GF-HVAC (250 A)	2	21.8	4	Partial	WARNING
INC-GF (500 A)	INC-DBGF1 (100 A)	5	21.8	4	Partial	WARNING
INC-FF (200 A)	FF-OFF (63 A)	3.17	14.3	1.6	Partial	WARNING
INC-FF (200 A)	FF-IT (40 A)	5	14.3	1.6	Partial	WARNING
INC-FF (200 A)	FF-LAB (80 A)	2.5	14.3	1.6	Partial	WARNING
INC-FF (200 A)	FF-LTG (25 A)	8	14.3	1.6	Partial	WARNING
INC-MCC (315 A)	MCC-WP (160 A)	1.97	19	2.5	Partial	WARNING
INC-MCC (315 A)	MCC-FP (125 A)	2.52	19	2.5	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-Y1 (6 A)	16.67	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-R1 (10 A)	10	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-Y2 (6 A)	16.67	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-B6 (10 A)	10	10.7	0.8	Partial	WARNING

Upstream	Downstream	In ratio	Fault at downstream (kA)	Selectivity limit (kA)	Short circuit	Result
INC-DBGF1 (100 A)	DB-GF1-Y5 (16 A)	6.25	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-B7 (10 A)	10	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-R3 (40 A)	2.5	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-B3 (20 A)	5	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-R6 (6 A)	16.67	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-B4 (16 A)	6.25	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-Y4 (16 A)	6.25	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-B5 (16 A)	6.25	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-R5 (10 A)	10	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-Y8 (6 A)	16.67	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-Y3 (20 A)	5	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-R4 (16 A)	6.25	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-B2 (6 A)	16.67	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-Y7 (6 A)	16.67	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-B1 (6 A)	16.67	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-R7 (6 A)	16.67	10.7	0.8	Partial	WARNING
INC-DBGF1 (100 A)	DB-GF1-Y6 (16 A)	6.25	10.7	0.8	Partial	WARNING

13. Earth fault loop impedance and disconnection

Scope: Whole installation (5 boards)

Overall status: **PASS** (36 pass · 0 warning · 0 fail)

13.1 Input

Earthing arrangement	TN-S
Supply loop (Ze)	From the source impedance to each board

13.2 Design criteria

Disconnection time	IEC 60364-4-41 Table 41.1: 0.4 s final circuits, 5 s distribution circuits
Fault current	Minimum, c = 0.95
Protective conductor	Adiabatic, k = 143

13.3 Method

- TN-S system. Earth fault loop impedance Zs = Ze (at the supply board) + (R phase + R cpc) of the circuit, at operating temperature; minimum fault current with c = 0.95.
- Protective conductor (adiabatic): $S \geq I \cdot \sqrt{t} \div 143$, t = 0.1 s when tripping instantaneously (not the device let-through energy), with I the current through the conductor checked. Parallel runs: each CPC carries an equal share of the end-of-circuit fault (identical runs bonded at both ends); a result that passes only on that assumption is marked "assumed sharing" (warning) because a fault within one run or unequal runs are not covered. The total fault current sets the disconnection check.
- Disconnection per IEC 60364-4-41 Table 41.1 (TN, U0 in the 230 V band): 0.4 s for final circuits up to 63 A with socket-outlets and up to 32 A supplying fixed equipment only; 5 s for distribution circuits and other final circuits. Circuits whose purpose is not set are treated as socket-outlet circuits. Instantaneous tripping when If ≥ Ia; below that, a 5 s circuit needs the breaker's time-current curve checked. Protective conductor checked by the adiabatic equation (k = 143).

13.4 Result

Circuits checked

36 pass · 0 warning · 0 fail

13.5 Verification — Earth fault loop on each board

Table 20

Board	Circuits	Highest Zs ÷ max Zs	Pass / check / fail	Status
MDB-1	5	74 %	5 / 0 / 0	PASS
SMDB-GF	4	70 %	4 / 0 / 0	PASS
DB-GF1	21	0 %	21 / 0 / 0	PASS
SMDB-FF	4	46 %	4 / 0 / 0	PASS
MCC-1	2	38 %	2 / 0 / 0	PASS

See also Appendix C (short-circuit and earth fault calculations).

14. Phase balance

Scope: Whole installation (5 boards)

Overall status: **WARNING** (3 pass · 2 warning · 0 fail)

14.1 Method

- Demand kW and kvar per phase (each load's demand factor and power factor), downstream boards included: single-phase circuits on their labelled phase R / Y / B; 3-phase loads spread evenly.
- Phase current I = S ÷ U0 at nominal voltage (U0 = 240 V). Neutral current = phasor sum of the three phase currents, phases 120° apart, at fundamental frequency — triplen harmonics (e.g. from LED drivers and IT loads) add to the neutral and are not included.
- Current unbalance = largest deviation from the average phase current ÷ the average. Flagged above 10 % (app default, not a standard's limit).
- Limits of this check: demand-based at nominal voltage — not an unbalanced load flow; no voltage unbalance, no neutral voltage, no phase-specific voltage drop. Single-phase circuits with no phase set are spread evenly, which hides their unbalance; they are listed.

14.2 Result

Boards checked
3 pass · 2 warning · 0 fail

Single-phase circuits without a phase
1 (FF-LTG) — spread evenly, unbalance may be understated

14.3 Phase balance at each board (demand)

Table 21

Board	R (A)	Y (A)	B (A)	R / Y / B (kW)	Neutral (A)	Unbalance	No phase set	Result
MDB-1	776.1	774.2	773	167.4 / 167 / 166.7	2.7	0.2 %	1	WARNING
SMDB-GF	326.8	324.9	323.7	68.1 / 67.7 / 67.5	2.7	0.5 %	0	PASS
DB-GF1	60.4	58.4	57.3	13 / 12.6 / 12.3	2.7	2.9 %	0	PASS
SMDB-FF	135.5	135.5	135.5	29 / 29 / 29	0	0 %	1	WARNING
MCC-1	210.3	210.3	210.3	43.3 / 43.3 / 43.3	0	0 %	0	PASS

15. Busbar trunking risers

Scope: Whole installation (5 boards)

15.1 Method

- Demand: the tap-offs' loads (a floor's board demand, or the load entered) × diversity (IEC 61439-6: 0.9 for 2–3 tap-offs, 0.8 for 4–5, 0.7 for 6–9, 0.6 for 10 or more, unless set).
- Rating: Ib ≤ In (feeding breaker) ≤ busway rating × ambient derating (≈ 1 % per °C above 40 °C).
- Voltage drop floor by floor: $\Delta U = \sqrt{3} \cdot I \cdot L \cdot (R \cos \phi + X \sin \phi)$, each riser section carrying the diversified load of the tap-offs above it.
- Short-circuit: the busway's Icw (1 s) ≥ the fault level at the board feeding it. Busway data: typical sandwich busway — confirm with the manufacturer.

15.2 Result

Risers checked

0 pass · 0 warning · 0 fail

15.3 Busbar risers

Nothing in scope.

16. Power factor correction

Scope: Whole installation (5 boards)

Overall status: **WARNING** (0 pass · 1 warning · 0 fail)

16.1 Method

- Strategy: Central — at the main board (MDB).
- Qc = P × (tan ϕ1 – tan ϕ2) to PF 0.95, from the maximum demand (demand factors applied). Capacitors already on the drawing are included.
- Banks are sized from the bottom up: each bank only covers what the banks below it do not, so nothing is corrected twice.
- Board banks: automatic (APFC) in 25 kvar steps, rounded up. Individual capacitors: fixed, rounded down (never over-correct a motor).
- Detuned (7 %, 189 Hz) where non-linear load ≥ 25 % of demand. Capacitor rating ≥ 1.05 × U ÷ (1 – p).
- Breaker ≥ 1.43 × bank current (IEC 60831 1.3 × overcurrent and capacitance tolerance). Light-load check at 30 % of demand.

16.2 Result

Target PF
0.95

Capacitors (new)
100 kvar in 1 banks

MDB-1 PF
0.88 → 0.954

MDB-1 capacity released
39 kVA (transformer 51 % → 47 %)

16.3 Capacitor banks

Table 22

Location	Demand (kW)	Reactive (kvar)	Existing (kvar)	Banks below (kvar)	PF now	Required (kvar)	Bank	Detuning · capacitor V	Breaker · cable	PF after	Current (A)	Result
MDB-1	451	243	—	—	0.88	94.2	100 kvar (4 × 25)	None · 440 V	200 A · 4C × 70 mm²	0.954	712 → 658	WARNING

16.4 Main boards

Table 23

Main board	Demand (kW)	PF before	PF after	kVA before → after	Released (kVA)	Transformer loading
MDB-1	451	0.88	0.954	512 → 473	39	51 % → 47 %

16.5 Notes

Table 24

Location	Note
MDB-1	17 % non-linear load — check harmonics; consider a detuned bank

17. DB and cable schedules

Scope: Whole installation (5 boards)

17.1 Method

- Schedules of the boards in scope, from the design data and the latest calculation run.

17.2 Result

Boards
5

See also Appendix A (detailed load schedule), Appendix B (cable calculations).

18. Results summary

18.1 Transformers

Table 25

Main board	Installed (kVA)	%Z	Required (kVA)	Demand (kVA)	Loading	Status
MDB-1	1000	5	710	473	47 %	PASS

18.2 Standby generator

Table 26

Essential demand (kVA)	Required (kVA)	Recommended	Installed (kVA)	Status
64	544	630 kVA	Not defined	DATA REQUIRED

18.3 Distribution boards (MDB, SMDB, DB, MCC)

Table 27

Board	Type	Bus rating (A)	Demand (A)	Loading	Ik″ (kA)	Status
MDB-1	MDB	1600	712	45 %	29.2	PASS
SMDB-GF	SMDB	630	325	52 %	21.8	PASS
DB-GF1	DB	100	59	59 %	10.7	PASS
SMDB-FF	SMDB	250	135	54 %	14.3	PASS
MCC-1	MCC	400	210	53 %	19	PASS

18.4 Capacitor banks

Table 28

Location	Bank (kvar)	Steps	PF before → after	Target PF	Status
MDB-1	100	4 × 25	0.88 → 0.954	0.95	WARNING

18.5 Load summary

Table 29

Item	Value
Connected load	514 kW
Maximum demand	451 kW · 512 kVA
Transformer MDB-1 loading	47 % of 1000 kVA (design limit 80 %)
Transformer MDB-1 spare	527 kVA
Generator loading	Installed rating not defined

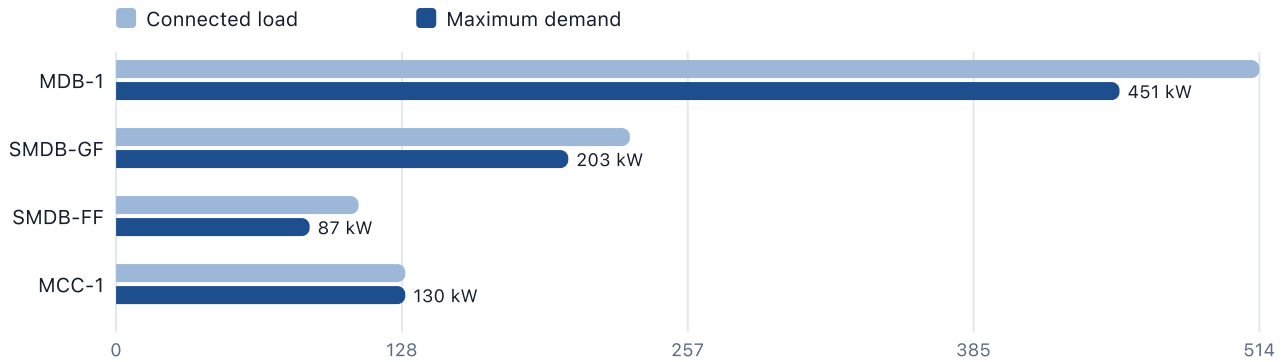


Figure 1 — Connected load vs maximum demand. Per board, kW. The gap is the diversity applied by the demand factors.

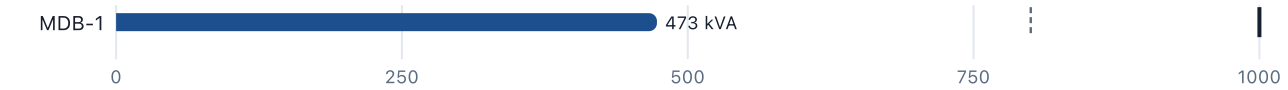


Figure 2 — Transformer capacity vs demand. Demand against the installed rating (tick) and the 80 % design loading limit (dashed).

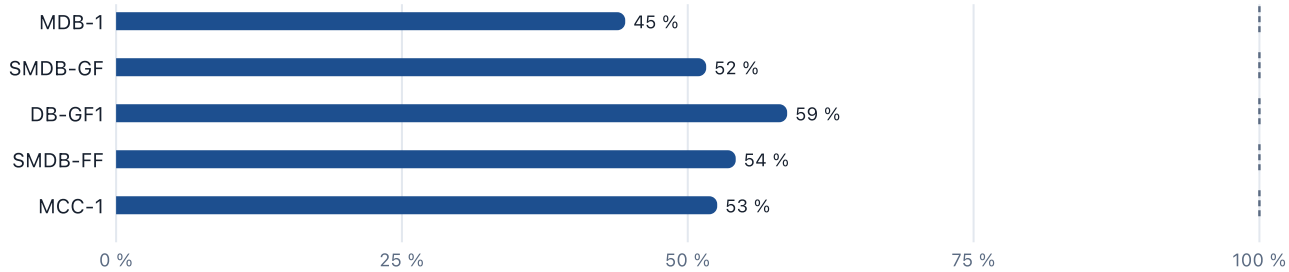


Figure 3 — Board loading. Demand current ÷ bus rating. Boards without a rating are not shown.

18.6 Cable summary

Table 30

From	To	Load (kW)	Current (A)	Cable	Protective device	Voltage drop (%)	Status
MDB-1	SMDB-GF	—	327	4C × 300 mm²	500 A MCCB	0.46	PASS
MDB-1	SMDB-FF	—	135	4C × 95 mm²	200 A MCCB	0.75	PASS
MDB-1	MCC-1	—	210	4C × 150 mm²	315 A MCCB	0.58	PASS
MDB-1	EV charging – parking	30.8	44	4C × 10 mm²	63 A C	2.38	PASS
MDB-1	Solar PV inverter	50	70	4C × 25 mm²	100 A MCCB	0.76	PASS
SMDB-GF	Lighting	40.5	63	4C × 16 mm²	80 A MCCB	1.47	PASS
SMDB-GF	Sockets	16.8	27	4C × 4 mm²	40 A C	2.37	PASS

From	To	Load (kW)	Current (A)	Cable	Protective device	Voltage drop (%)	Status
SMDB-GF	HVAC	108	177	4C × 95 mm²	250 A MCCB	1.16	PASS
SMDB-FF	Office	28	43	4C × 10 mm²	63 A C	1.66	PASS
SMDB-FF	IT room	20	29	4C × 4 mm²	40 A C	2.04	PASS
SMDB-FF	Lab	35	57	4C × 16 mm²	80 A MCCB	1.62	PASS
SMDB-FF	Corridor lighting	4	19	2C × 6 mm²	25 A C	2.57	PASS
MCC-1	Water pump	75	121	4C × 50 mm²	160 A MCCB	1.03	PASS
MCC-1	Fire pump	55	89	4C × 35 mm²	125 A MCCB	1.12	PASS
SMDB-GF	DB-GF1	—	60	4C × 35 mm²	100 A MCCB	0.92	PASS

Final circuits are on the DB load schedules.

19. Compliance summary

Overall design status: **WARNING** (8 PASS · 1 WARNING · 1 DATA REQUIRED)

Table 31

Design check	Governing item	Calculated	Requirement / selected	Margin	Status
Transformer loading	MDB-1	473 kVA	1000 kVA	527 kVA	PASS
Generator loading	Standby generator	64 kVA	Installed rating not defined	—	DATA REQUIRED
Fault level (breaking capacity)	GF-LTG	21.8 kA	25 kA	3.2 kA	PASS
Voltage drop	DB-GF1-R3	2.79 %	4 %	1.21 %	PASS
Cable capacity (Ib ≤ Iz)	FF-IT	29 A	41 A	12 A	PASS
Overload protection (Ib ≤ In ≤ Iz)	36 circuits	36 of 36 meet it	All circuits	—	PASS
Board loading	DB-GF1	59 A	100 A	41 A	PASS
Earth fault disconnection	DB-EV	Zs 0.266 Ω	≤ 0.361 Ω	0.096 Ω	PASS
Selectivity	31 breaker pairs	0 total, 31 partial / to check	Total selectivity	—	WARNING
Power factor	MDB-1	0.954	≥ 0.95	0.004	PASS

Each row shows the item with the least margin; the status is the worst over all items of that check. PASS / WARNING / FAIL are the app's calculated checks against the project limits; NOT CHECKED: nothing installed or selected to check against; DATA REQUIRED: a rating or input is missing. Engineering review and approval are separate.

Appendix A — Detailed load schedule

A.1 DB and cable schedules — DB schedule

Table 32

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
MDB-1	1	INC-GF	Incomer to Ground floor SMDB	Sub-board SMDB-GF					327	500 A MCCB	TP+N	36	4C × 300 mm² XLPE/SWA + 150 mm² CPC	35	0.46	PASS	
MDB-1	2	INC-FF	Incomer to First floor SMDB	Sub-board SMDB-FF					135	200 A MCCB	TP+N	36	4C × 95 mm² XLPE/SWA + 50 mm² CPC	55	0.75	PASS	
MDB-1	3	INC-MCC	Incomer to Pump room MCC	Sub-board MCC-1					210	315 A MCCB	TP+N	36	4C × 150 mm² XLPE/SWA + 95 mm² CPC	40	0.58	PASS	
MDB-1	4	DB-EV	EV charging – parking	EV charging	44	0.7	30.8	0.98	44	63 A C	TP+N	36	4C × 10 mm² XLPE/SWA + 10 mm² CPC	60	2.38	PASS	
MDB-1	5	DB-PV	Solar PV inverter	Solar PV	50	1	50	1	70	100 A MCCB	TP+N	36	4C × 25 mm² XLPE/SWA + 16 mm² CPC	30	0.76	PASS	
MDB-1		TOTAL	Main Distribution Board — 1600 A		514		451.1	0.88	712			29.2				45% loaded	
SMDB-GF	1	GF-LTG	Lighting	Lighting	45	0.9	40.5	0.9	63	80 A MCCB	TP+N	25	4C × 16 mm² XLPE/SWA + 16 mm² CPC	30	1.47	PASS	
SMDB-GF	2	GF-SKT	Sockets	Small power / sockets	28	0.6	16.8	0.85	27	40 A C	TP+N	25	4C × 4 mm² XLPE/SWA + 4 mm² CPC	35	2.37	PASS	
SMDB-GF	3	GF-HVAC	HVAC	HVAC	120	0.9	108	0.85	177	250 A MCCB	TP+N	25	4C × 95 mm² XLPE/SWA + 50 mm² CPC	40	1.16	PASS	
SMDB-GF	4	INC-DBGF1	Incomer to Villa ground floor DB	Sub-board DB-GF1					60	100 A MCCB	TP+N	25	4C × 35 mm² XLPE/SWA + 16 mm² CPC	30	0.92	PASS	
SMDB-GF		TOTAL	Ground floor SMDB — 630 A		231		203.3	0.87	325			21.8				52% loaded	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
DB-GF1	1	DB-GF1-Y1	Majlis lighting	Lighting	1	1	1	0.9	5	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	18	1.48	PASS	
DB-GF1	2	DB-GF1-R1	Living lighting	Lighting	1.2	1	1.2	0.9	6	10 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	20	1.67	PASS	
DB-GF1	3	DB-GF1-Y2	Kitchen lighting	Lighting	0.8	1	0.8	0.9	4	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	22	1.49	PASS	
DB-GF1	4	DB-GF1-B6	Majlis sockets	Small power / sockets	1.5	1	1.5	0.9	7	10 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	18	1.44	PASS	
DB-GF1	5	DB-GF1-Y5	Living sockets	Small power / sockets	2	1	2	0.9	9	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	20	1.7	PASS	
DB-GF1	6	DB-GF1-B7	Kitchen sockets	Small power / sockets	1.3	1	1.3	0.9	6	10 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	22	1.45	PASS	
DB-GF1	7	DB-GF1-R3	Cooker	General	6	1	6	0.9	28	40 A C	SP+N	15	2C × 6 mm² XLPE/SWA + 6 mm² CPC	24	2.79	PASS	
DB-GF1	8	DB-GF1-B3	Kitchen water heater	General	3	1	3	0.9	14	20 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	24	2.32	PASS	
DB-GF1	9	DB-GF1-R6	Maid room	Small power / sockets	1.1	1	1.1	0.9	5	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	28	1.51	PASS	
DB-GF1	10	DB-GF1-B4	Majlis split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	18	1.79	PASS	
DB-GF1	11	DB-GF1-Y4	Living split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	20	1.89	PASS	
DB-GF1	12	DB-GF1-B5	Dining split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	22	1.99	PASS	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
DB-GF1	13	DB-GF1-R5	Guest bedroom	Small power / sockets	1.5	1	1.5	0.9	7	10 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	26	1.67	PASS	
DB-GF1	14	DB-GF1-Y8	Guest bathroom	Lighting	0.3	1	0.3	0.9	1	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	27	1.05	PASS	
DB-GF1	15	DB-GF1-Y3	Guest bath water heater	General	3	1	3	0.9	14	20 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	27	2.49	PASS	
DB-GF1	16	DB-GF1-R4	Guest bedroom split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	26	2.18	PASS	
DB-GF1	17	DB-GF1-B2	Dining lighting	Lighting	0.6	1	0.6	0.9	3	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	22	1.33	PASS	
DB-GF1	18	DB-GF1-Y7	Dining sockets	Small power / sockets	1	1	1	0.9	5	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	22	1.35	PASS	
DB-GF1	19	DB-GF1-B1	External lighting	Lighting	1	1	1	0.9	5	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	35	2.01	PASS	
DB-GF1	20	DB-GF1-R7	Water pump	Motor	0.8	1	0.8	0.9	3	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	30	1.36	PASS	
DB-GF1	21	DB-GF1-Y6	Laundry	Small power / sockets	2	1	2	0.9	9	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	25	1.89	PASS	
DB-GF1		TOTAL	Villa ground floor DB — 100 A		38		38	0.9	59			10.7				59% loaded	
SMDB-FF	1	FF-OFF	Office	Small power / sockets	35	0.8	28	0.9	43	63 A C	TP+N	25	4C × 10 mm² XLPE/SWA + 10 mm² CPC	25	1.66	PASS	
SMDB-FF	2	FF-IT	IT room	IT / data	20	1	20	0.95	29	40 A C	TP+N	25	4C × 4 mm² XLPE/SWA + 4 mm² CPC	20	2.04	PASS	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
SMDB-FF	3	FF-LAB	Lab	General	50	0.7	35	0.85	57	80 A MCCB	TP+N	25	4C × 16 mm² XLPE/SWA + 16 mm² CPC	30	1.62	PASS	
SMDB-FF	4	FF-LTG	Corridor lighting	Lighting	4	1	4	0.9	19	25 A C	SP+N	25	2C × 6 mm² XLPE/SWA + 6 mm² CPC	35	2.57	PASS	
SMDB-FF		TOTAL	First floor SMDB — 250 A		109		87	0.89	135			14.3				54% loaded	
MCC-1	1	MCC-WP	Water pump	Motor	75	1	75	0.86	121	160 A MCCB	TP+N	25	4C × 50 mm² XLPE/SWA + 25 mm² CPC	20	1.03	PASS	
MCC-1	2	MCC-FP	Fire pump	Fire pump	55	1	55	0.86	89	125 A MCCB	TP+N	25	4C × 35 mm² FR + 16 mm² CPC	25	1.12	PASS	
MCC-1		TOTAL	Pump room MCC — 400 A		130		130	0.86	210			19				53% loaded	

Appendix B — Cable calculations

B.1 Cable and breaker sizing — Cables and breakers

Table 33

Circuit	From	To	Cable	Installation	Derating	Ib (A)	Breaker In (A)	Iz (A)	Ib ≤ In ≤ Iz	ΔV total (%)	Result
INC-GF	MDB-1	SMDB-GF	4C × 300 mm² XLPE/SWA + 150 CPC	E	0.87 · 1.00	326.8	500 MCCB	525	PASS	0.46	PASS
INC-FF	MDB-1	SMDB-FF	4C × 95 mm² XLPE/SWA + 50 CPC	E	0.87 · 1.00	135.5	200 MCCB	266	PASS	0.75	PASS
INC-MCC	MDB-1	MCC-1	4C × 150 mm² XLPE/SWA + 95 CPC	E	0.87 · 1.00	210.3	315 MCCB	342	PASS	0.58	PASS
DB-EV	MDB-1	EV charging – parking	4C × 10 mm² XLPE/SWA + 10 CPC	E	0.87 · 1.00	43.7	63 C	70	PASS	2.38	PASS
DB-PV	MDB-1	Solar PV inverter	4C × 25 mm² XLPE/SWA + 16 CPC	E	0.87 · 1.00	69.6	100 MCCB	120	PASS	0.76	PASS
GF-LTG	SMDB-GF	Lighting	4C × 16 mm² XLPE/SWA + 16 CPC	E	0.87 · 1.00	62.6	80 MCCB	91	PASS	1.47	PASS
GF-SKT	SMDB-GF	Sockets	4C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	27.5	40 C	41	PASS	2.37	PASS
GF-HVAC	SMDB-GF	HVAC	4C × 95 mm² XLPE/SWA + 50 CPC	E	0.87 · 1.00	176.8	250 MCCB	266	PASS	1.16	PASS
FF-OFF	SMDB-FF	Office	4C × 10 mm² XLPE/SWA + 10 CPC	E	0.87 · 1.00	43.3	63 C	70	PASS	1.66	PASS
FF-IT	SMDB-FF	IT room	4C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	29.3	40 C	41	PASS	2.04	PASS
FF-LAB	SMDB-FF	Lab	4C × 16 mm² XLPE/SWA + 16 CPC	E	0.87 · 1.00	57.3	80 MCCB	91	PASS	1.62	PASS
FF-LTG	SMDB-FF	Corridor lighting	2C × 6 mm² XLPE/SWA + 6 CPC	E	0.87 · 1.00	18.5	25 C	52	PASS	2.57	PASS
MCC-WP	MCC-1	Water pump	4C × 50 mm² XLPE/SWA + 25 CPC	E	0.87 · 1.00	121.3	160 MCCB	174	PASS	1.03	PASS
MCC-FP	MCC-1	Fire pump	4C × 35 mm² FR + 16 CPC	E	0.87 · 1.00	89	125 MCCB	146	PASS	1.12	PASS
INC-DBGF1	SMDB-GF	DB-GF1	4C × 35 mm² XLPE/SWA + 16 CPC	E	0.87 · 1.00	60.4	100 MCCB	146	PASS	0.92	PASS
DB-GF1-Y1	DB-GF1	Majlis lighting	2C × 2.5 mm² XLPE/SWA + 2.5 CPC	E	0.87 · 1.00	4.6	6 C	31	PASS	1.48	PASS
DB-GF1-R1	DB-GF1	Living lighting	2C × 2.5 mm² XLPE/SWA + 2.5 CPC	E	0.87 · 1.00	5.6	10 C	31	PASS	1.67	PASS
DB-GF1-Y2	DB-GF1	Kitchen lighting	2C × 2.5 mm² XLPE/SWA + 2.5 CPC	E	0.87 · 1.00	3.9	6 C	31	PASS	1.49	PASS
DB-GF1-B6	DB-GF1	Majlis sockets	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	7	10 C	41	PASS	1.44	PASS
DB-GF1-Y5	DB-GF1	Living sockets	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	9.3	16 C	41	PASS	1.7	PASS
DB-GF1-B7	DB-GF1	Kitchen sockets	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	5.8	10 C	41	PASS	1.45	PASS
DB-GF1-R3	DB-GF1	Cooker	2C × 6 mm² XLPE/SWA + 6 CPC	E	0.87 · 1.00	27.8	40 C	52	PASS	2.79	PASS
DB-GF1-B3	DB-GF1	Kitchen water heater	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	13.9	20 C	41	PASS	2.32	PASS
DB-GF1-R6	DB-GF1	Maid room	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	5.1	6 C	41	PASS	1.51	PASS
DB-GF1-B4	DB-GF1	Majlis split A/C	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	11.6	16 C	41	PASS	1.79	PASS
DB-GF1-Y4	DB-GF1	Living split A/C	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	11.6	16 C	41	PASS	1.89	PASS
DB-GF1-B5	DB-GF1	Dining split A/C	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	11.6	16 C	41	PASS	1.99	PASS

Circuit	From	To	Cable	Installation	Derating	Ib (A)	Breaker In (A)	Iz (A)	Ib ≤ In ≤ Iz	ΔV total (%)	Result
DB-GF1-R5	DB-GF1	Guest bedroom	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	6.9	10 C	41	PASS	1.67	PASS
DB-GF1-Y8	DB-GF1	Guest bathroom	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	1.2	6 C	41	PASS	1.05	PASS
DB-GF1-Y3	DB-GF1	Guest bath water heater	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	13.9	20 C	41	PASS	2.49	PASS
DB-GF1-R4	DB-GF1	Guest bedroom split A/C	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	11.6	16 C	41	PASS	2.18	PASS
DB-GF1-B2	DB-GF1	Dining lighting	2C × 2.5 mm² XLPE/SWA + 2.5 CPC	E	0.87 · 1.00	2.8	6 C	31	PASS	1.33	PASS
DB-GF1-Y7	DB-GF1	Dining sockets	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	4.6	6 C	41	PASS	1.35	PASS
DB-GF1-B1	DB-GF1	External lighting	2C × 2.5 mm² XLPE/SWA + 2.5 CPC	E	0.87 · 1.00	4.6	6 C	31	PASS	2.01	PASS
DB-GF1-R7	DB-GF1	Water pump	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	3.5	6 C	41	PASS	1.36	PASS
DB-GF1-Y6	DB-GF1	Laundry	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	9.3	16 C	41	PASS	1.89	PASS

B.2 Load flow and voltage drop study — Feeders

Table 34

Circuit	From	To	Cable	Length (m)	Ib (A)	PF	ΔV cable (%)	ΔV total (%)	Result
INC-GF	MDB-1	SMDB-GF	4C × 300 mm²	35	326.8	0.87	0.46	0.46	PASS
INC-FF	MDB-1	SMDB-FF	4C × 95 mm²	55	135.5	0.89	0.75	0.75	PASS
INC-MCC	MDB-1	MCC-1	4C × 150 mm²	40	210.3	0.86	0.58	0.58	PASS
DB-EV	MDB-1	EV charging – parking	4C × 10 mm²	60	43.7	0.98	2.38	2.38	PASS
DB-PV	MDB-1	Solar PV inverter	4C × 25 mm²	30	69.6	1	0.76	0.76	PASS
GF-LTG	SMDB-GF	Lighting	4C × 16 mm²	30	62.6	0.9	1	1.47	PASS
GF-SKT	SMDB-GF	Sockets	4C × 4 mm²	35	27.5	0.85	1.91	2.37	PASS
GF-HVAC	SMDB-GF	HVAC	4C × 95 mm²	40	176.8	0.85	0.7	1.16	PASS
FF-OFF	SMDB-FF	Office	4C × 10 mm²	25	43.3	0.9	0.91	1.66	PASS
FF-IT	SMDB-FF	IT room	4C × 4 mm²	20	29.3	0.95	1.29	2.04	PASS
FF-LAB	SMDB-FF	Lab	4C × 16 mm²	30	57.3	0.85	0.87	1.62	PASS
FF-LTG	SMDB-FF	Corridor lighting	2C × 6 mm²	35	18.5	0.9	1.82	2.57	PASS
MCC-WP	MCC-1	Water pump	4C × 50 mm²	20	121.3	0.86	0.45	1.03	PASS
MCC-FP	MCC-1	Fire pump	4C × 35 mm²	25	89	0.86	0.54	1.12	PASS
INC-DBGF1	SMDB-GF	DB-GF1	4C × 35 mm²	30	60.4	0.9	0.45	0.92	PASS

B.3 DB and cable schedules — Cable schedule

Table 35

Cable tag	From	To	Ref	Type	Cores × size	CPC (mm²)	Length (m)	Tray route	Ib (A)	Breaker In (A)	Iz (A)	Vd cable (%)	Vd total (%)	Status
C-INC-GF	MDB-1	SMDB-GF	1	Cu/XLPE/SWA/PVC (armoured)	4C × 300 mm²	150	35		327	500	525	0.46	0.46	PASS
C-INC-FF	MDB-1	SMDB-FF	6	Cu/XLPE/SWA/PVC (armoured)	4C × 95 mm²	50	55		135	200	266	0.75	0.75	PASS
C-INC-MCC	MDB-1	MCC-1	4	Cu/XLPE/SWA/PVC (armoured)	4C × 150 mm²	95	40		210	315	342	0.58	0.58	PASS
C-DB-EV	MDB-1	EV charging – parking	12	Cu/XLPE/SWA/PVC (armoured)	4C × 10 mm²	10	60		44	63	70	2.38	2.38	PASS
C-DB-PV	MDB-1	Solar PV inverter	10	Cu/XLPE/SWA/PVC (armoured)	4C × 25 mm²	16	30		70	100	120	0.76	0.76	PASS
C-GF-LTG	SMDB-GF	Lighting	11	Cu/XLPE/SWA/PVC (armoured)	4C × 16 mm²	16	30		63	80	91	1	1.47	PASS
C-GF-SKT	SMDB-GF	Sockets	14	Cu/XLPE/SWA/PVC (armoured)	4C × 4 mm²	4	35		27	40	41	1.91	2.37	PASS
C-GF-HVAC	SMDB-GF	HVAC	6	Cu/XLPE/SWA/PVC (armoured)	4C × 95 mm²	50	40		177	250	266	0.7	1.16	PASS
C-INC-DBGF1	SMDB-GF	DB-GF1	9	Cu/XLPE/SWA/PVC (armoured)	4C × 35 mm²	16	30		60	100	146	0.45	0.92	PASS
C-DB-GF1-Y1	DB-GF1	Majlis lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	18		5	6	31	0.56	1.48	PASS
C-DB-GF1-R1	DB-GF1	Living lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	20		6	10	31	0.75	1.67	PASS
C-DB-GF1-Y2	DB-GF1	Kitchen lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	22		4	6	31	0.58	1.49	PASS
C-DB-GF1-B6	DB-GF1	Majlis sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	18		7	10	41	0.52	1.44	PASS
C-DB-GF1-Y5	DB-GF1	Living sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	20		9	16	41	0.78	1.7	PASS
C-DB-GF1-B7	DB-GF1	Kitchen sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	22		6	10	41	0.53	1.45	PASS
C-DB-GF1-R3	DB-GF1	Cooker	25	Cu/XLPE/SWA/PVC (armoured)	2C × 6 mm²	6	24		28	40	52	1.88	2.79	PASS
C-DB-GF1-B3	DB-GF1	Kitchen water heater	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	24		14	20	41	1.4	2.32	PASS

Cable tag	From	To	Ref	Type	Cores × size	CPC (mm²)	Length (m)	Tray route	Ib (A)	Breaker In (A)	Iz (A)	Vd cable (%)	Vd total (%)	Status
C-DB-GF1-R6	DB-GF1	Maid room	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	28		5	6	41	0.59	1.51	PASS
C-DB-GF1-B4	DB-GF1	Majlis split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	18		12	16	41	0.87	1.79	PASS
C-DB-GF1-Y4	DB-GF1	Living split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	20		12	16	41	0.97	1.89	PASS
C-DB-GF1-B5	DB-GF1	Dining split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	22		12	16	41	1.07	1.99	PASS
C-DB-GF1-R5	DB-GF1	Guest bedroom	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	26		7	10	41	0.75	1.67	PASS
C-DB-GF1-Y8	DB-GF1	Guest bathroom	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	27		1	6	41	0.14	1.05	PASS
C-DB-GF1-Y3	DB-GF1	Guest bath water heater	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	27		14	20	41	1.57	2.49	PASS
C-DB-GF1-R4	DB-GF1	Guest bedroom split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	26		12	16	41	1.26	2.18	PASS
C-DB-GF1-B2	DB-GF1	Dining lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	22		3	6	31	0.41	1.33	PASS
C-DB-GF1-Y7	DB-GF1	Dining sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	22		5	6	41	0.43	1.35	PASS
C-DB-GF1-B1	DB-GF1	External lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	35		5	6	31	1.09	2.01	PASS
C-DB-GF1-R7	DB-GF1	Water pump	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	30		3	6	41	0.44	1.36	PASS
C-DB-GF1-Y6	DB-GF1	Laundry	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	25		9	16	41	0.97	1.89	PASS
C-FF-OFF	SMDB-FF	Office	12	Cu/XLPE/SWA/PVC (armoured)	4C × 10 mm²	10	25		43	63	70	0.91	1.66	PASS
C-FF-IT	SMDB-FF	IT room	14	Cu/XLPE/SWA/PVC (armoured)	4C × 4 mm²	4	20		29	40	41	1.29	2.04	PASS
C-FF-LAB	SMDB-FF	Lab	11	Cu/XLPE/SWA/PVC (armoured)	4C × 16 mm²	16	30		57	80	91	0.87	1.62	PASS
C-FF-LTG	SMDB-FF	Corridor lighting	25	Cu/XLPE/SWA/PVC (armoured)	2C × 6 mm²	6	35		19	25	52	1.82	2.57	PASS

Cable tag	From	To	Ref	Type	Cores × size	CPC (mm²)	Length (m)	Tray route	Ib (A)	Breaker In (A)	Iz (A)	Vd cable (%)	Vd total (%)	Status
C-MCC-WP	MCC-1	Water pump	8	Cu/XLPE/SWA/PVC (armoured)	4C × 50 mm²	25	20		121	160	174	0.45	1.03	PASS
C-MCC-FP	MCC-1	Fire pump	9 *	Fire-rated, BS 8491 (120 min, large cables)	4C × 35 mm²	16	25		89	125	146	0.54	1.12	PASS

Appendix C — Short-circuit and earth fault calculations

C.1 Short circuit study — Breaker breaking capacity

Table 36

Circuit	Board	To	Breaker	Ik″ at breaker (kA)	Icu (kA)	Margin	Result
INC-GF	MDB-1	SMDB-GF	500 A MCCB	29.2	36	23 %	PASS
INC-FF	MDB-1	SMDB-FF	200 A MCCB	29.2	36	23 %	PASS
INC-MCC	MDB-1	MCC-1	315 A MCCB	29.2	36	23 %	PASS
DB-EV	MDB-1	EV charging – parking	63 A C	29.2	36	23 %	PASS
DB-PV	MDB-1	Solar PV inverter	100 A MCCB	29.2	36	23 %	PASS
GF-LTG	SMDB-GF	Lighting	80 A MCCB	21.8	25	15 %	PASS
GF-SKT	SMDB-GF	Sockets	40 A C	21.8	25	15 %	PASS
GF-HVAC	SMDB-GF	HVAC	250 A MCCB	21.8	25	15 %	PASS
FF-OFF	SMDB-FF	Office	63 A C	14.3	25	75 %	PASS
FF-IT	SMDB-FF	IT room	40 A C	14.3	25	75 %	PASS
FF-LAB	SMDB-FF	Lab	80 A MCCB	14.3	25	75 %	PASS
FF-LTG	SMDB-FF	Corridor lighting	25 A C	14.3	25	75 %	PASS
MCC-WP	MCC-1	Water pump	160 A MCCB	19	25	32 %	PASS
MCC-FP	MCC-1	Fire pump	125 A MCCB	19	25	32 %	PASS
INC-DBGF1	SMDB-GF	DB-GF1	100 A MCCB	21.8	25	15 %	PASS
DB-GF1-Y1	DB-GF1	Majlis lighting	6 A C	10.7	15	41 %	PASS
DB-GF1-R1	DB-GF1	Living lighting	10 A C	10.7	15	41 %	PASS
DB-GF1-Y2	DB-GF1	Kitchen lighting	6 A C	10.7	15	41 %	PASS
DB-GF1-B6	DB-GF1	Majlis sockets	10 A C	10.7	15	41 %	PASS
DB-GF1-Y5	DB-GF1	Living sockets	16 A C	10.7	15	41 %	PASS
DB-GF1-B7	DB-GF1	Kitchen sockets	10 A C	10.7	15	41 %	PASS
DB-GF1-R3	DB-GF1	Cooker	40 A C	10.7	15	41 %	PASS
DB-GF1-B3	DB-GF1	Kitchen water heater	20 A C	10.7	15	41 %	PASS
DB-GF1-R6	DB-GF1	Maid room	6 A C	10.7	15	41 %	PASS
DB-GF1-B4	DB-GF1	Majlis split A/C	16 A C	10.7	15	41 %	PASS
DB-GF1-Y4	DB-GF1	Living split A/C	16 A C	10.7	15	41 %	PASS
DB-GF1-B5	DB-GF1	Dining split A/C	16 A C	10.7	15	41 %	PASS

Circuit	Board	To	Breaker	Ik″ at breaker (kA)	Icu (kA)	Margin	Result
DB-GF1-R5	DB-GF1	Guest bedroom	10 A C	10.7	15	41 %	PASS
DB-GF1-Y8	DB-GF1	Guest bathroom	6 A C	10.7	15	41 %	PASS
DB-GF1-Y3	DB-GF1	Guest bath water heater	20 A C	10.7	15	41 %	PASS
DB-GF1-R4	DB-GF1	Guest bedroom split A/C	16 A C	10.7	15	41 %	PASS
DB-GF1-B2	DB-GF1	Dining lighting	6 A C	10.7	15	41 %	PASS
DB-GF1-Y7	DB-GF1	Dining sockets	6 A C	10.7	15	41 %	PASS
DB-GF1-B1	DB-GF1	External lighting	6 A C	10.7	15	41 %	PASS
DB-GF1-R7	DB-GF1	Water pump	6 A C	10.7	15	41 %	PASS
DB-GF1-Y6	DB-GF1	Laundry	16 A C	10.7	15	41 %	PASS

C.2 Short circuit study — Fault at the end of each cable

Table 37

Circuit	From	To	Cable	Length (m)	Ik at cable end (kA)
INC-GF	MDB-1	SMDB-GF	4C × 300 mm²	35	21.81
INC-FF	MDB-1	SMDB-FF	4C × 95 mm²	55	14.3
INC-MCC	MDB-1	MCC-1	4C × 150 mm²	40	19.01
DB-EV	MDB-1	EV charging – parking	4C × 10 mm²	60	2.24
DB-PV	MDB-1	Solar PV inverter	4C × 25 mm²	30	9.7
GF-LTG	SMDB-GF	Lighting	4C × 16 mm²	30	6.2
GF-SKT	SMDB-GF	Sockets	4C × 4 mm²	35	1.52
GF-HVAC	SMDB-GF	HVAC	4C × 95 mm²	40	13.94
FF-OFF	SMDB-FF	Office	4C × 10 mm²	25	4.2
FF-IT	SMDB-FF	IT room	4C × 4 mm²	20	2.38
FF-LAB	SMDB-FF	Lab	4C × 16 mm²	30	5.11
FF-LTG	SMDB-FF	Corridor lighting	2C × 6 mm²	35	1.1
MCC-WP	MCC-1	Water pump	4C × 50 mm²	20	12.95
MCC-FP	MCC-1	Fire pump	4C × 35 mm²	25	10.53
INC-DBGF1	SMDB-GF	DB-GF1	4C × 35 mm²	30	10.66
DB-GF1-Y1	DB-GF1	Majlis lighting	2C × 2.5 mm²	18	0.88
DB-GF1-R1	DB-GF1	Living lighting	2C × 2.5 mm²	20	0.8
DB-GF1-Y2	DB-GF1	Kitchen lighting	2C × 2.5 mm²	22	0.73

Circuit	From	To	Cable	Length (m)	Ik at cable end (kA)
DB-GF1-B6	DB-GF1	Majlis sockets	2C × 4 mm²	18	1.35
DB-GF1-Y5	DB-GF1	Living sockets	2C × 4 mm²	20	1.23
DB-GF1-B7	DB-GF1	Kitchen sockets	2C × 4 mm²	22	1.13
DB-GF1-R3	DB-GF1	Cooker	2C × 6 mm²	24	1.5
DB-GF1-B3	DB-GF1	Kitchen water heater	2C × 4 mm²	24	1.04
DB-GF1-R6	DB-GF1	Maid room	2C × 4 mm²	28	0.9
DB-GF1-B4	DB-GF1	Majlis split A/C	2C × 4 mm²	18	1.35
DB-GF1-Y4	DB-GF1	Living split A/C	2C × 4 mm²	20	1.23
DB-GF1-B5	DB-GF1	Dining split A/C	2C × 4 mm²	22	1.13
DB-GF1-R5	DB-GF1	Guest bedroom	2C × 4 mm²	26	0.97
DB-GF1-Y8	DB-GF1	Guest bathroom	2C × 4 mm²	27	0.94
DB-GF1-Y3	DB-GF1	Guest bath water heater	2C × 4 mm²	27	0.94
DB-GF1-R4	DB-GF1	Guest bedroom split A/C	2C × 4 mm²	26	0.97
DB-GF1-B2	DB-GF1	Dining lighting	2C × 2.5 mm²	22	0.73
DB-GF1-Y7	DB-GF1	Dining sockets	2C × 4 mm²	22	1.13
DB-GF1-B1	DB-GF1	External lighting	2C × 2.5 mm²	35	0.47
DB-GF1-R7	DB-GF1	Water pump	2C × 4 mm²	30	0.85
DB-GF1-Y6	DB-GF1	Laundry	2C × 4 mm²	25	1

C.3 Earth fault loop impedance and disconnection — Earth fault loop

Table 38

Circuit	Board	Breaker	CPC (mm²)	Zs (Ω)	Max Zs (Ω)	If (A)	Ia (A)	Required (s)	Disconnection	CPC min, each (mm²)	Result
INC-GF	MDB-1	500 A MCCB	150	0.0144	0.0455	15804	5000	5	< 0.1 s	34.9	PASS
INC-FF	MDB-1	200 A MCCB	50	0.0419	0.1138	5433	2000	5	< 0.1 s	12	PASS
INC-MCC	MDB-1	315 A MCCB	95	0.0204	0.0723	11150	3150	5	< 0.1 s	24.7	PASS
DB-EV	MDB-1	63 A C	10	0.2656	0.3613	857	630	0.4	< 0.1 s	1.9	PASS
DB-PV	MDB-1	100 A MCCB	16	0.0701	0.2276	3246	1000	5	< 0.1 s	7.2	PASS
GF-LTG	SMDB-GF	80 A MCCB	16	0.0932	0.2845	2442	800	5	< 0.1 s	5.4	PASS
GF-SKT	SMDB-GF	40 A C	4	0.3969	0.5691	573	400	0.4	< 0.1 s	1.3	PASS
GF-HVAC	SMDB-GF	250 A MCCB	50	0.0398	0.091	5724	2500	5	< 0.1 s	12.7	PASS
FF-OFF	SMDB-FF	63 A C	10	0.1505	0.3613	1512	630	0.4	< 0.1 s	3.3	PASS

Circuit	Board	Breaker	CPC (mm²)	Zs (Ω)	Max Zs (Ω)	If (A)	Ia (A)	Required (s)	Disconnection	CPC min, each (mm²)	Result
FF-IT	SMDB-FF	40 A C	4	0.2616	0.5691	870	400	0.4	< 0.1 s	1.9	PASS
FF-LAB	SMDB-FF	80 A MCCB	16	0.1237	0.2845	1840	800	5	< 0.1 s	4.1	PASS
FF-LTG	SMDB-FF	25 A C	6	0.2991	0.9105	761	250	0.4	< 0.1 s	1.7	PASS
MCC-WP	MCC-1	160 A MCCB	25	0.0455	0.1423	4997	1600	5	< 0.1 s	11.1	PASS
MCC-FP	MCC-1	125 A MCCB	16	0.0685	0.1821	3325	1250	5	< 0.1 s	7.4	PASS
INC-DBGF1	SMDB-GF	100 A MCCB	16	0.0709	0.2276	3209	1000	5	< 0.1 s	7.1	PASS
DB-GF1-Y1	DB-GF1	6 A C	2.5	0.3901	455.2407	584	1	0.4	< 0.1 s	1.3	PASS
DB-GF1-R1	DB-GF1	10 A C	2.5	0.4256	455.2407	535	1	0.4	< 0.1 s	1.2	PASS
DB-GF1-Y2	DB-GF1	6 A C	2.5	0.4612	455.2407	494	1	0.4	< 0.1 s	1.1	PASS
DB-GF1-B6	DB-GF1	10 A C	4	0.2693	1517.469	845	0	0.4	< 0.1 s	1.9	PASS
DB-GF1-Y5	DB-GF1	16 A C	4	0.2914	1517.469	781	0	0.4	< 0.1 s	1.7	PASS
DB-GF1-B7	DB-GF1	10 A C	4	0.3135	1517.469	726	0	0.4	< 0.1 s	1.6	PASS
DB-GF1-R3	DB-GF1	40 A C	6	0.2476	1517.469	919	0	0.4	< 0.1 s	2	PASS
DB-GF1-B3	DB-GF1	20 A C	4	0.3356	1517.469	678	0	0.4	< 0.1 s	1.5	PASS
DB-GF1-R6	DB-GF1	6 A C	4	0.3798	1517.469	599	0	0.4	< 0.1 s	1.3	PASS
DB-GF1-B4	DB-GF1	16 A C	4	0.2693	1517.469	845	0	0.4	< 0.1 s	1.9	PASS
DB-GF1-Y4	DB-GF1	16 A C	4	0.2914	1517.469	781	0	0.4	< 0.1 s	1.7	PASS
DB-GF1-B5	DB-GF1	16 A C	4	0.3135	1517.469	726	0	0.4	< 0.1 s	1.6	PASS
DB-GF1-R5	DB-GF1	10 A C	4	0.3577	1517.469	636	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-Y8	DB-GF1	6 A C	4	0.3687	1517.469	617	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-Y3	DB-GF1	20 A C	4	0.3687	1517.469	617	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-R4	DB-GF1	16 A C	4	0.3577	1517.469	636	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-B2	DB-GF1	6 A C	2.5	0.4612	455.2407	494	1	0.4	< 0.1 s	1.1	PASS
DB-GF1-Y7	DB-GF1	6 A C	4	0.3135	1517.469	726	0	0.4	< 0.1 s	1.6	PASS
DB-GF1-B1	DB-GF1	6 A C	2.5	0.6923	455.2407	329	1	0.4	< 0.1 s	0.7	PASS
DB-GF1-R7	DB-GF1	6 A C	4	0.4019	1517.469	566	0	0.4	< 0.1 s	1.3	PASS
DB-GF1-Y6	DB-GF1	16 A C	4	0.3466	1517.469	657	0	0.4	< 0.1 s	1.5	PASS

Appendix D — Equipment data

D.1 Transformer and standby generator sizing — Standby generator — loads

Table 39

Board / circuit	Share	Demand (kW)	Reactive (kvar)
MCC-FP Fire pump (essential circuit)	100 %	55	—

D.2 Boards

Table 40

Board	Type	Rating (A)	Busbar	IP	Manufacturer	Model	Location
MDB-1	MDB	1600	copper	IP42	Schneider Electric	PrismaSeT P	Main building – ground floor
SMDB-GF	SMDB	630	copper	IP42	Not defined	Not defined	Ground floor electrical room
SMDB-FF	SMDB	250	copper	IP42	Not defined	Not defined	First floor electrical room
MCC-1	MCC	400	copper	IP54	Not defined	Not defined	Basement pump room
DB-GF1	DB	100	copper	IP42	Not defined	Not defined	Villa GF – store room

D.3 Transformers

Table 41

Reference	Main board	Rating (kVA)	%Z	X/R	Vector group	Substation
MDB-1	MDB-1	1000	5	5 (app default)	Dyn11 (app default)	Not defined

Appendix E — Reference documents

E.1 Calculation basis

Table 42

Standard	Used for
IEC 60909-0	Short-circuit currents (simplified: c factor, infinite MV network)
IEC 60364-5-52	Cable current-carrying capacity, derating and voltage drop
IEC 60364-4-43	Overload protection: Ib ≤ In ≤ Iz
IEC 60364-4-41	Automatic disconnection: earth fault loop impedance and disconnection times
IEC 60364-5-54	Protective conductor size (Table 54.2 / adiabatic)
IEC 60898-1 / IEC 60947-2	Circuit-breaker characteristics and breaking capacity
IEC 60831	Capacitor bank rating and switching device
IEC 61439-6	Busbar trunking ratings